

UNIT 2

AGILE PROJECT DOCUMENTATION

Project Title

Enterprise Student Record and Academic Management Platform

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1. Introduction

The Enterprise Student Record and Academic Management Platform is designed to manage student academic information in a centralized and organized system.

Unit 2 focuses on applying Agile and Scrum methodologies to the development of the platform. The project is divided into smaller deliverable features called Epics and User Stories. These features are prioritized in a Product Backlog and implemented through Sprints.

Agile development enables continuous feedback, flexibility, faster delivery, and continuous improvement throughout the project.

2. Agile Product Vision

Product Vision Statement

To develop a centralized, secure, and user-friendly academic management platform that enables students, faculty, and administrators to efficiently manage student records, courses, attendance, examinations, marks, and academic results.

Vision Goals

- **Centralize student academic information.**
- **Reduce manual academic record management.**
- **Provide role-based access to users.**
- **Improve accuracy of student records.**
- **Provide faster access to academic information.**
- **Enable efficient course, attendance, and examination management.**
- **Support continuous improvement through Agile development.**

3. Scrum Team Structure

The project follows the Scrum framework.

Scrum Team

Role	Responsibility
Product Owner	Defines product requirements and prioritizes the backlog
Scrum Master	Facilitates Scrum activities and removes obstacles
Development Team	Designs, develops, tests, and integrates the system

Stakeholders

- **Students**
- **Faculty Members**
- **Academic Administrators**
- **Examination Cell Staff**
- **IT Support Team**

4. Scrum Roles and Responsibilities

Product Owner

- **Defines product vision.**
- **Identifies user requirements.**
- **Creates and prioritizes the Product Backlog.**
- **Communicates with stakeholders.**
- **Accepts or rejects completed features.**

Scrum Master

- **Organizes Scrum ceremonies.**
- **Ensures Scrum practices are followed.**
- **Removes development blockers.**
- **Supports communication within the team.**
- **Helps the team improve continuously.**

Development Team

- **Analyzes requirements.**

- **Designs system components.**
- **Develops application features.**
- **Performs testing.**
- **Fixes defects.**
- **Integrates completed features.**

5. Project Epics

The major features of the system are divided into the following Epics.

Epic 1: User Authentication and Access Control

Provides secure login and role-based access for students, faculty, administrators, and other authorized users.

Epic 2: Student Record Management

Allows administrators to create, update, search, and maintain student academic records.

Epic 3: Course Registration

Allows students to view available courses and register for required courses.

Epic 4: Attendance Management

Allows faculty members to record attendance and students to view their attendance information.

Epic 5: Examination and Marks Management

Allows examination staff and faculty to manage examinations, marks, and assessments.

Epic 6: Results and Academic Reports

Provides students and administrators with academic results and reports.

Epic 7: Role-Based Dashboards

Provides customized dashboards according to the user's role.

6. User Stories

Epic 1: Authentication

US01:

As a student, I want to log in securely so that I can access my academic information.

US02:

As an administrator, I want role-based access so that users can access only authorized features.

Epic 2: Student Records

US03:

As an administrator, I want to add student records so that student information is maintained digitally.

US04:

As an administrator, I want to update student information so that records remain accurate.

US05:

As an administrator, I want to search student records so that I can quickly find required information.

Epic 3: Course Registration

US06:

As a student, I want to view available courses so that I can select suitable courses.

US07:

As a student, I want to register for courses so that my academic registration is recorded.

Epic 4: Attendance

US08:

As a faculty member, I want to record student attendance so that attendance data is maintained.

US09:

As a student, I want to view my attendance so that I can monitor my attendance percentage.

Epic 5: Examination and Marks

US10:

As examination staff, I want to create examination records so that examinations can be managed systematically.

US11:

As a faculty member, I want to enter student marks so that assessment results can be maintained.

Epic 6: Results and Reports

US12:

As a student, I want to view my examination results so that I can track my academic

performance.

US13:

As an administrator, I want to generate academic reports so that student performance can be analyzed.

Epic 7: Dashboards

US14:

As a user, I want a role-based dashboard so that I can quickly access relevant information and functions.

7.Acceptance Criteria

User Story	Acceptance Criteria
US01	User can enter valid credentials and access the system
US02	Users can access only features permitted for their role
US03	Administrator can add a new student record successfully
US04	Administrator can edit and save updated student information
US05	Administrator can search students using relevant details
US06	Student can view available courses
US07	Student can select and register for available courses
US08	Faculty can record attendance for students
US09	Student can view current attendance information
US10	Examination staff can create examination

	records
US11	Faculty can enter and save marks
US12	Student can view published examination results
US13	Administrator can generate academic reports
US14	Dashboard displays features according to user role

8. Product Backlog

ID	Product Backlog Item	Priority	Story Points
US01	Secure Login	High	3
US02	Role-Based Access	High	5
US03	Add Student Record	High	5
US04	Update Student Record	High	3
US05	Search Student Records	Medium	3
US06	View Available Courses	High	3
US07	Course Registration	High	5
US08	Record Attendance	High	5
US09	View Attendance	Medium	3

US10	Examination Management	High	5
US11	Marks Management	High	5
US12	View Results	High	3
US13	Academic Reports	Medium	5
US14	Role-Based Dashboard	Medium	5

9. Story Point Estimation

Story points are used to estimate the relative complexity and effort required for each User Story.

The project uses the following scale:

1, 2, 3, 5, 8

Estimation Guidelines

- 1–2 points: Very simple task
- 3 points: Small task with limited complexity
- 5 points: Medium-complexity task
- 8 points: Large or highly complex task

Story points consider:

- Complexity
- Development effort
- Testing effort
- Dependencies
- Technical uncertainty

10. Sprint Planning

The project is divided into short development cycles called Sprints.

Each Sprint focuses on delivering a working set of features.

Sprint Duration

2 Weeks per Sprint

Sprint Objectives

Sprint 1

Focus on system foundation and authentication.

Features:

- **Secure Login**
- **Role-Based Access**
- **Basic Dashboard**

Sprint 2

Focus on student record and course management.

Features:

- **Add Student Record**
- **Update Student Record**
- **Search Student Records**
- **View Available Courses**
- **Course Registration**

Sprint 3

Focus on attendance and examination management.

Features:

- **Record Attendance**
- **View Attendance**
- **Examination Management**
- **Marks Management**

Sprint 4

Focus on results, reports, testing, and final integration.

Features:

- **View Results**
- **Academic Reports**
- **Final Dashboard Improvements**
- **System Testing**
- **Bug Fixing**

11. Sprint Backlog

Sprint 1 Backlog

ID	Task	Story Points
US01	Secure Login	3
US02	Role-Based Access	5
US14	Basic Role-Based Dashboard	5

Total: 13 Story Points

Sprint 2 Backlog

ID	Task	Story Points
US03	Add Student Record	5
US04	Update Student Record	3
US05	Search Student Records	3
US06	View Available Courses	3
US07	Course Registration	5

Total: 19 Story Points

Sprint 3 Backlog

ID	Task	Story Points
US08	Record Attendance	5

US09	View Attendance	3
US10	Examination Management	5
US11	Marks Management	5

Total: 18 Story Points

Sprint 4 Backlog

ID	Task	Story Points
US12	View Results	3
US13	Academic Reports	5
US14	Dashboard Improvements	5
-	System Testing and Bug Fixing	5

Total: 18 Story Points

12. Scrum Board

The Scrum Board is used to track the progress of tasks during a Sprint.

To Do	In Progress	Testing	Done
Student Records	Course Registration	Login Testing	Project Setup
Attendance	Dashboard	Course Testing	Requirements
Reports	Marks Management	Attendance Testing	UI Prototype
Results	-	-	-

The board helps the team visualize work and identify tasks that are blocked or delayed.

13. Burndown Chart

A Burndown Chart represents the amount of remaining work during a Sprint.

Example Sprint Burndown Data

Sprint Day	Remaining Story Points
Day 1	13
Day 3	11
Day 5	9
Day 7	6
Day 9	3
Day 10	0

The expected trend is a gradual reduction in remaining work until all Sprint tasks are completed.

14. Sprint Review

At the end of every Sprint, the completed work is demonstrated to stakeholders.

Sprint 1 Review

Completed:

- Login functionality
- Role-based access
- Initial dashboard

Sprint 2 Review

Completed:

- Student record management
- Student search

- **Course viewing**
- **Course registration**

Sprint 3 Review

Completed:

- **Attendance management**
- **Examination management**
- **Marks entry**

Sprint 4 Review

Completed:

- **Results**
- **Academic reports**
- **Dashboard improvements**
- **Final testing and integration**

Stakeholder feedback is collected and used to update the Product Backlog.

15. Sprint Retrospective

The Sprint Retrospective is conducted after each Sprint to identify improvements.

What Went Well

- **Clear division of tasks.**
- **Regular communication between team members.**
- **Incremental feature development.**
- **Continuous testing of completed features.**

What Could Be Improved

- **Improve estimation accuracy.**
- **Reduce task dependencies.**
- **Improve documentation consistency.**
- **Identify technical issues earlier.**

Action Items

- **Conduct better Sprint Planning.**
- **Break large tasks into smaller tasks.**
- **Perform testing throughout the Sprint.**
- **Maintain regular communication.**
- **Review backlog priorities regularly.**

16. Agile Framework Comparison

Feature	Scrum	Kanban	XP
Work Cycle	Fixed Sprints	Continuous	Iterative
Roles	Clearly defined	Flexible	Defined technical roles
Planning	Sprint Planning	Continuous	Frequent planning
Best For	Product development	Continuous workflow	Software quality
Progress Tracking	Scrum Board/Burndown	Kanban Board	Engineering practices
Project Choice	Selected	Alternative	Alternative

Selected Framework

Scrum is selected for this project because the platform can be developed incrementally through defined Sprints with clear roles, backlog management, Sprint Reviews, and Retrospectives.

17. Project Timeline and Release Roadmap

Phase	Duration	Major Deliverables
Sprint 1	Week 1-2	Authentication and Dashboard
Sprint 2	Week 3-4	Student Records and Course Registration
Sprint 3	Week 5-6	Attendance and Examination Management
Sprint 4	Week 7-8	Results, Reports, Testing

Release Roadmap

Release 1 – Foundation

- **Login**
- **Authentication**
- **Role-Based Access**
- **Dashboard**

Release 2 – Academic Management

- **Student Records**
- **Course Registration**
- **Attendance**

Release 3 – Examination Management

- **Examination Management**
- **Marks Management**
- **Results**

Release 4 – Final Product

- **Academic Reports**
- **Testing**
- **Bug Fixing**
- **Final Integration**

18. Agile Development Benefits

Agile development provides the following benefits for this project:

- **Faster delivery of features.**
- **Continuous stakeholder feedback.**
- **Flexible response to changing requirements.**
- **Early identification of defects.**
- **Incremental development.**
- **Better team collaboration.**
- **Continuous product improvement.**

19. Final Deliverables

The Unit 2 Agile project documentation includes:

- ✓ **Agile Product Vision**
- ✓ **Scrum Team Structure**
- ✓ **Scrum Roles and Responsibilities**
- ✓ **Project Epics**
- ✓ **User Stories**
- ✓ **Acceptance Criteria**
- ✓ **Product Backlog**
- ✓ **Story Point Estimation**
- ✓ **Sprint Planning**
- ✓ **Sprint Backlog**
- ✓ **Scrum Board**
- ✓ **Burndown Chart**
- ✓ **Sprint Review**
- ✓ **Sprint Retrospective**
- ✓ **Agile Framework Comparison**
- ✓ **Project Timeline and Release Roadmap**

20. Conclusion

The Agile and Scrum methodology provides a structured approach for developing the Enterprise Student Record and Academic Management Platform.

By dividing the project into Epics, User Stories, Product Backlog items, and Sprints, the development team can deliver the platform incrementally while continuously incorporating stakeholder feedback.

The Scrum framework also promotes collaboration, transparency, continuous testing, and improvement, making it suitable for the development of this academic management platform.